Tool Execution Finished: tool, params, raw_output
Resource Resolver: Extract Canonical Accesses (db:*, file:*, api:*, git:*)
Access Mode?
read
write / delete
Bind Access to Current Resource Version v_curr Record Read Access in Access Table
Increment Version: v_new = v_curr + 1 Update Resource Record
Find Prior Writers (v < v_new): Mark SUPERSEDED (W = 0.0, taint = 0.0) Add SUPERSEDES edge in DAG
Find Prior Readers (v_access < v_new): Mark STALE (W = 0.5 × taint) Add INVALIDATES edge in DAG
Collect Seed Invalidation Nodes (Super + Stale Nodes)
BFS Propagation along DEPENDS_ON Edges: For each consumer node C: taint_C = taint_parent × gamma If taint_C >= floor: Mark SUSPECT
Truth Maintenance / Recompute: Re-execution of node resets taint = 1.0 and heals valid path
Done: Node Marked Fresh (taint = 1.0)